

Learning Backward Compatible Representations

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Abstract

In today's multimedia-rich environment, the rapid growth of data poses significant challenges for developing efficient multi-modal retrieval systems essential for retrieving text, images, audio, and video. As data expands, newer, scalable, and high-performance retrieval systems are increasingly necessary. Embedding-based deep neural networks (DNNs) have become key solutions, transforming high-dimensional data into lower-dimensional embeddings for easy comparison and retrieval. However, updating DNNs changes the internal feature representations, necessitating the extraction of new feature vectors for all gallery data, which is costly, especially with gallery sets comprising billions of data. Learning backward-compatible representations addresses this by allowing new representation to be matched with old gallery data without recalculating features. This tutorial aims to equip participants with the knowledge and tools to apply backward-compatible representations, enhancing multimedia retrieval systems' efficiency and scalability. Participants will learn the importance of compatible representations, basic methods and techniques, and explore challenging open questions that are becoming increasingly relevant to multimedia and cross-modal retrieval.¹

CCS Concepts

• **Computing methodologies** → **Artificial intelligence; Learning paradigms.**

Keywords

Deep Learning, Representation Learning, Compatible Learning, Fixed Classifiers, Neural Collapse.

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1 Motivation and Impact on the Multimedia Community

1.1 Subject

In today's multimedia-rich environment, the rapid growth of data poses significant challenges for developing efficient multi-modal retrieval systems. These systems are essential for retrieving relevant data such as text, images, audio, and video. As data continues to expand, scalable and high-performance retrieval systems are increasingly necessary, driving advancements in technology.

Embedding-based deep neural networks (DNNs) have become a key solution, transforming high-dimensional data into lower-dimensional embeddings that capture semantic meanings and enable easy comparison and retrieval. However, a major challenge arises when a DNN is updated, as the internal representations change, complicating the update process for retrieval systems. This necessitates re-indexing, which involves re-extracting new feature representations for all the data in the gallery set.

Learning backward-compatible representations has emerged as a crucial solution to this problem. It allows for the development of new representation models without the need to recalculate features for the gallery data. Re-indexing can be computationally unsustainable for large gallery sets and may even be unfeasible if the original data is no longer available due to privacy concerns or storage restrictions. Backward compatible training addresses this issue by training new models to produce embeddings that can be directly compared with old representations, allowing the new model to process user queries that are matched with the old gallery features.

Through this tutorial, participants will gain:

- (1) A comprehensive understanding of why learning compatible representations is and will be fundamental in the coming years.
- (2) An introduction to the basic methods and techniques for learning backward-compatible representations and their applications, ranging from continual learning to multi-modal learning and LLMs applications.
- (3) Insight into challenging open questions for solving real-world applications in multi-modal retrieval systems.

To achieve these goals, the lecture will:

- Explain the limitations of the current practices in retrieval systems and provide the foundation and motivation for learning backward-compatible representations.
- Use the problem of unimodal search as a cornerstone example to illustrate the effectiveness of existing methods.
- Expand the discussion to scenarios where the model is updated using only new data (continual learning), multi-modal areas, and the current practice of leveraging third-party pre-trained models.
- Provide an overview of key questions and open challenges in implementing backward-compatible cross-modal retrieval systems.

1.2 Why This Tutorial is Relevant for the Multimedia Community

Research on Backward-Compatible Learning began in 2020 with seminal work by Amazon and has been gaining increasing attention in recent years, with numerous publications in top-tier journals and conferences. While most research has focused on unimodal scenarios, the challenge of learning compatible representations extends to the multimedia domain and the growing trend of enhancing Large Language Models (LLMs) with retrieval systems, such as Retrieval-Augmented Generation (RAG).

This tutorial aims to equip participants with the knowledge and tools to apply backward-compatible representations, thereby enhancing the efficiency and scalability of multimedia retrieval systems. Although the primary focus is on unimodal scenarios, the concepts discussed are also relevant to multimedia and cross-modal retrieval, benefiting a broad audience.

2 Tentative Schedule and Lecture Topics

A half-day tutorial is proposed consisting of three main sessions. The first session introduces the basic principles and methodologies of learning backward-compatible representations. The audience will learn about the motivations and algorithms for achieving compatibility of representations across multiple model updates. The second session delves into learning backward-compatible representations within lifelong learning scenarios. This includes exploring challenges when old training data is unavailable during model updates and discussing methods to maintain compatibility while incorporating new data. The third session focuses on learning backward-compatible representations in the era of multi-modal large language models (LLMs). This includes discussing backward-compatible cross-modal representations and the challenges associated with multi-modal learning and Retrieval-Augmented Generation (RAG) systems. Finally, the tutorial will showcase real-world applications and potential new directions for backward-compatible representations in the multimedia community.

2.1 Tentative Schedule

- (1) **Learning Backward-Compatible Representations**
 - (a) **Standard Approach For Retrieval Systems.** Discuss the standard approach in retrieval, which involves training on a static dataset.

- (b) **Why Retrieval Systems Need to be Updated.** Explain the necessity of updating internal representations in retrieval systems and introduce the problem addressed by backward-compatible representations.
- (c) **Introduction to Backward-Compatible Learning Representations.** Provide motivations and definitions, and explain the basic concepts of backward-compatible learning representations.
- (d) **Methods and Techniques for Backward-Compatible Representation Learning.** Review different approaches for achieving backward compatibility, including techniques that help new models work well with old ones. Discuss practical examples and real-world applications.
- (e) **Q&A and Discussion.**
- (2) **Learning Backward-Compatible Representations within Lifelong Learning Scenarios**
 - (a) **Lifelong Learning and Backward-Compatible Representations.** Explore challenges when old training data is unavailable during model updates. Focus on updating the model with new data while maintaining compatibility with old data.
 - (b) **Techniques for Lifelong Learning Representations.** Discuss methods for achieving compatibility in lifelong learning scenarios and explore practical applications and case studies.
 - (c) **Compatible Learning with Large Pretrained Models.** Discuss the trend in AI towards using pre-trained models and fine-tuning them for specific tasks. Present the challenge of maintaining compatibility when replacing a pre-trained model with a new one.
 - (d) **Q&A and Discussion.**
 - (3) **Learning Backward-Compatible Representations in the Era of Multi-modal LLMs**
 - (a) **Backward-Compatible Cross-Modal Representations.** Discuss the challenges and techniques for achieving backward compatibility in cross-modal retrieval systems.
 - (b) **Backward-Compatible Representations with RAG Systems.** Explore new challenges in systems using Large Language Models (LLMs) with Retrieval-Augmented Generation.
 - (c) **Open Challenges.** Highlight the open questions and future directions for research in this area.
 - (d) **Q&A and Discussion.**

2.2 Prerequisites and Supporting Material

The course is at the intermediate level which requires the audience to have some basic knowledge in deep learning, representation learning, continual (or lifelong) learning and retrieval system. The discussion about the applications of learning backward-compatible embedding will be self-contained, and past research experience in this specific domain is not crucial for the audience. The lecture material (notes and slides) will be made available immediately during the tutorial. The course will be supplemented by hands-on programming demonstration using publicly available code in the PyTorch framework. Expected Number of Attendees: 20-30.